ADMS

16.25.1

Model Export Installation Guide

Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
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1. Introduction

The Model Export tool is designed to export the "as-operated" distribution network model from a real-time ADMS into the Common Information Model (CIM) format, JSON format, or CYMDIST format for the Eaton's CYME software. The exported data can be integrated into third-party planning environments or other applications. This enables utilities to take advantage of the DMS data transformation and incremental model build process across other systems.

The CIM profile used by the Model Export tool is CIMDNOMProfile, which is an GE custom eCIM profile developed from the base CIM 15 model.

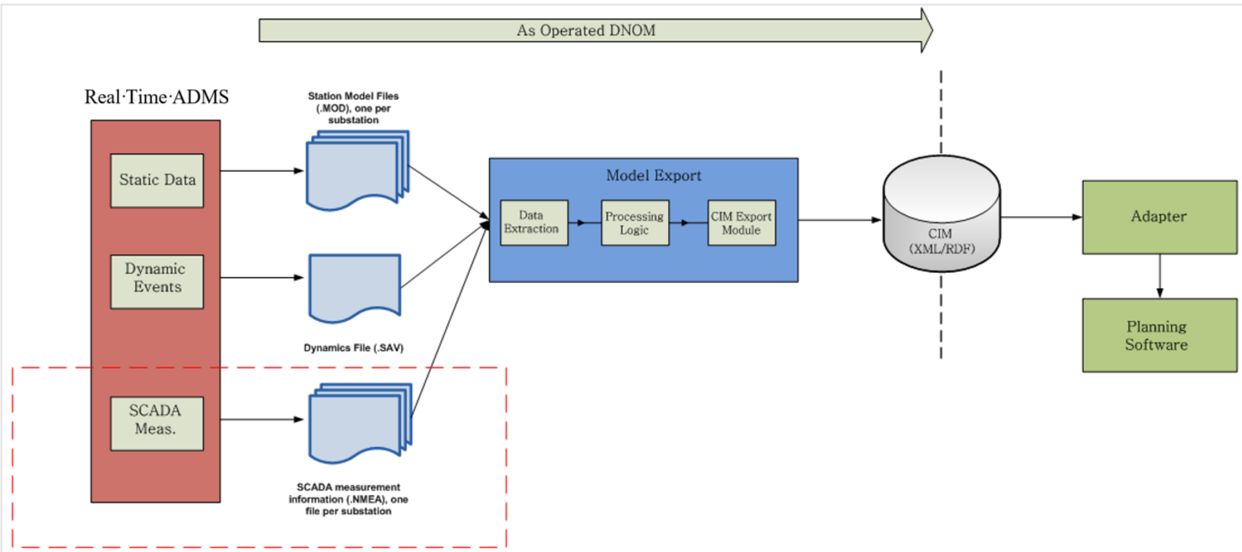


Figure 1. Model Export Workflow

Model Export can extract data from a DMS savecase (in `tar.gz` format) or directly from the model, measurement, or dynamics files.

The tool extracts data from the station models in the CIM, CYMDIST, or JSON file. For example, data may include:

- Substation (ID, Name, Alternative Name, Address, Placements, etc.)
- Line (Wire type, Wire arrangement and cable info types, Conductor type, Sizing, Ampacity, Construction type, Construction model name, etc.)
- Switch (Rated voltage, Rated current, Nameplate details, Current position, etc.)
- Transformer (Primary nominal voltage, Secondary nominal voltage, Primary rated voltage, Secondary rated voltage, Primary winding type, Secondary winding type, Primary tap position, Secondary tap position, etc.)

2. Installing Model Export

This section describes how to install the Model Export tool.

The Model Export software is packaged in the `ADMS_modelexport_X64.msi` installation kit. The installation kit provides shortcuts for the tool in the Windows Start menu.

> **Caution**
>
> Prior to installing an upgrade, you may want to preserve the previous version's configuration file, `ModelExportConfig.XML`.

2.1. Prerequisites

For a list of supported platforms and third-party applications, refer to the *ADMS Release Notes*.

No additional prerequisites are necessary to install Model Export.

2.2. Uninstalling an Earlier Version

To uninstall Model Export:

1. From the Windows Start menu, click Control Panel, and open Programs and Features.
2. Uninstall ADMS Model Export (X64).

2.3. Installing Model Export

To install Model Export, run the `ADMS_modelexport_X64.msi` installer and follow the instructions in the installation wizard.

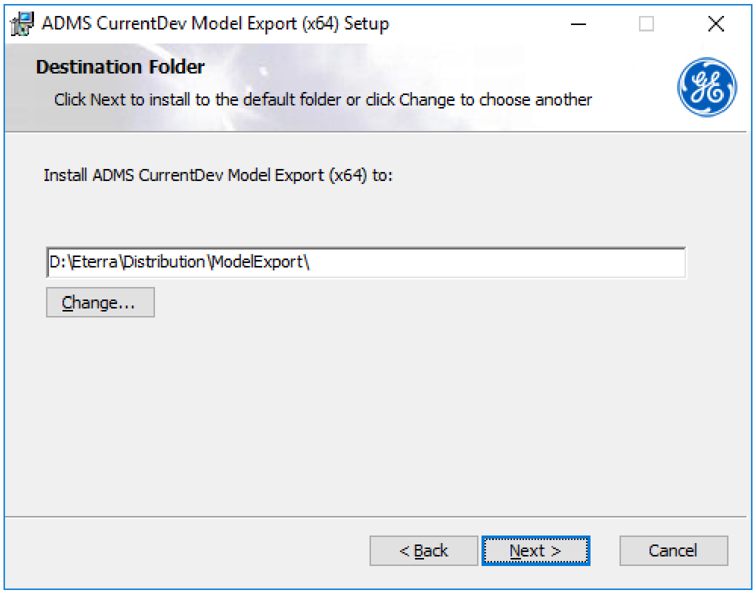


Figure 2. Installing Model Export

Verify the installation by opening the Model Export shortcut, as described in Launching the Model Export.

2.3.1. Side-By-Side Installation

The Model Export kit supports installing several instances of the same product on the same machine. For example, you can have Model Export tools for both ADMS 3.8.6 and ADMS 3.10, or two instances of Model Export tools for ADMS 3.10. Up to 25 instances of the kit can be installed.

To install an additional instance of Model Export:

1. Open a command prompt.
2. Navigate to the folder that contains the Model Export kit to be installed.
3. Run the following command:

   ```
   msiexec MSINEWINSTANCE=1 /i <Kit_Name> transforms=:i<Instance_Index> instancename=<Instance_Name>
   ```

   Where:

   - **Kit_Name:** Name or full path to the Model Export installation kit.
   - **Instance_Index:** Unique transform index of the instance. Can be 1 to 25.
   - **Instance_Name:** Unique instance name used in the installation path and product name.

     For example:

     ```
     msiexec MSINEWINSTANCE=1 /i ADMS_modelexport_x64.msi transforms=:i2 instancename=3.7
     ```

By default, new instances of the Model Export tool are installed in

`D:\Eterra\Distribution\<Instance_Name>\ModelExport\`.

2.4. Model Export Configuration File

The configuration file, `ModelExportConfig.XML`, is installed in the following location by default:

```
D:\Eterra\Distribution\ModelExport\Bin
```

All edits to the configuration file are made using the Model Export tool.

After making any configuration changes, verify that all the paths in the XML file are resolvable and reachable by the LocalService account. For example, if %IDMS_ROOT% is in the path, it needs to be a system-scoped environment variable. Otherwise, replace %IDMS_ROOT% in the configuration file with corresponding absolute paths.

2.5. Configuring the Windows Shortcut

The Model Export program should start in the same location as its executable. The Model Export's shortcut passes a configuration file to the application. The shortcut is located in the Windows Start menu at Eterra > Distribution > Model Export > Model Export. This shortcut allows you to start the Model Export.

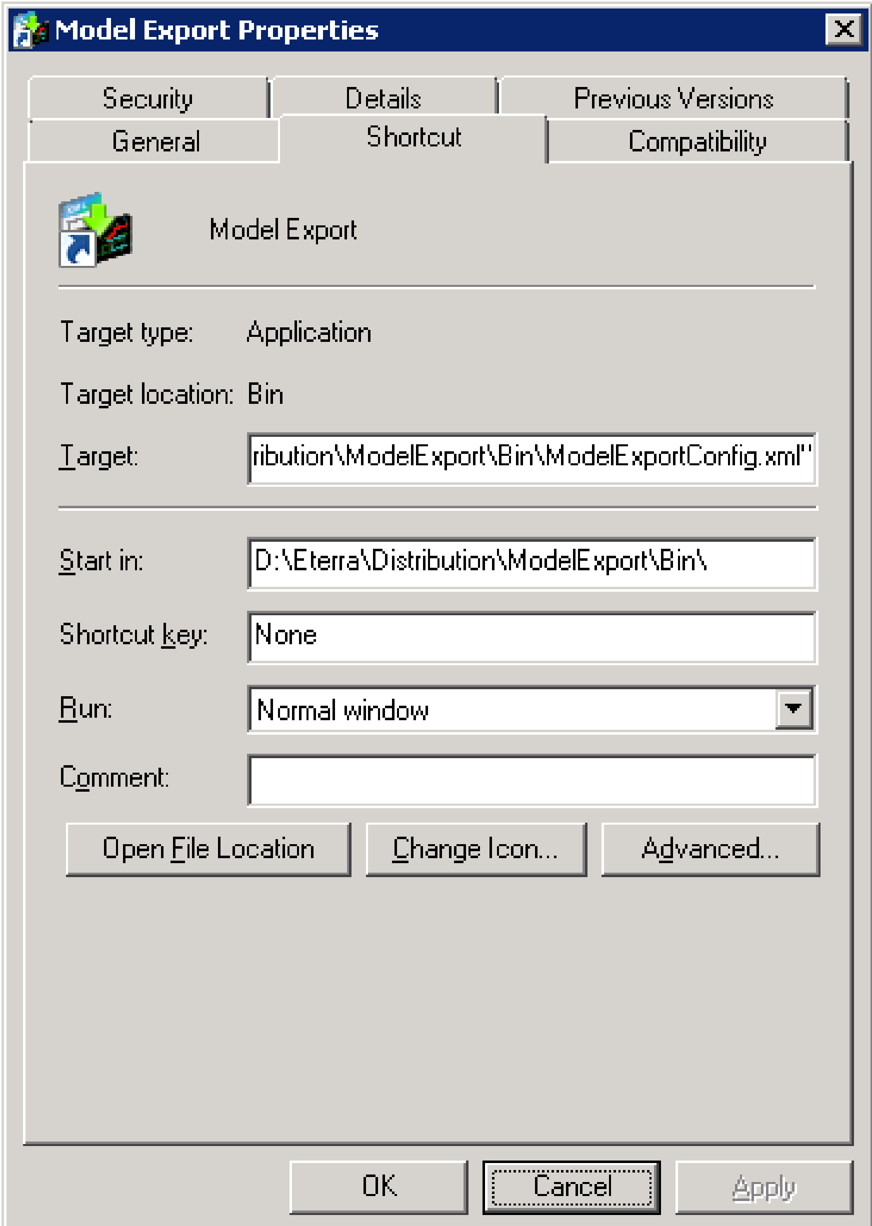


Figure 3. Model Export Shortcut Properties

You can modify the startup parameters via the Target and Start In fields of the shortcut's properties dialog box:

- Target = D:\Eterra\Distribution\ModelExport\Bin\ModelExport.exe /CONFIG "D:\Eterra\Distribution\ModelExport\Bin\ModelExportConfig.xml"
- Start in = D:\Eterra\Distribution\ModelExport\Bin\

> **Tip**
>
> You can use environment variables in the shortcut.

3. Model Export Displays Overview

This chapter describes the Model Export views as well as their fields and controls.

3.1. Model Export Controls

In Model Export, the File menu includes the following commands:

- **Load Config File:** Opens the Select Configuration File dialog box to select an existing configuration file.
- **Save Config File:** Saves the current configuration file.
- **Save As Config File:** Saves the current configuration file with a different name.
- **Exit:** Closes the application.

The Settings menu includes the following command:

- **Configuration:** Opens the Configuration dialog box to configure Model Export.

The Help menu includes the following command:

- **About Model Export:** Shows the version and copyright details for the tool.

3.2. General Tab

The General tab shows the list of station model files from the input folder. You can select stations and export their data to the CIM, CYMDIST, or JSON format.

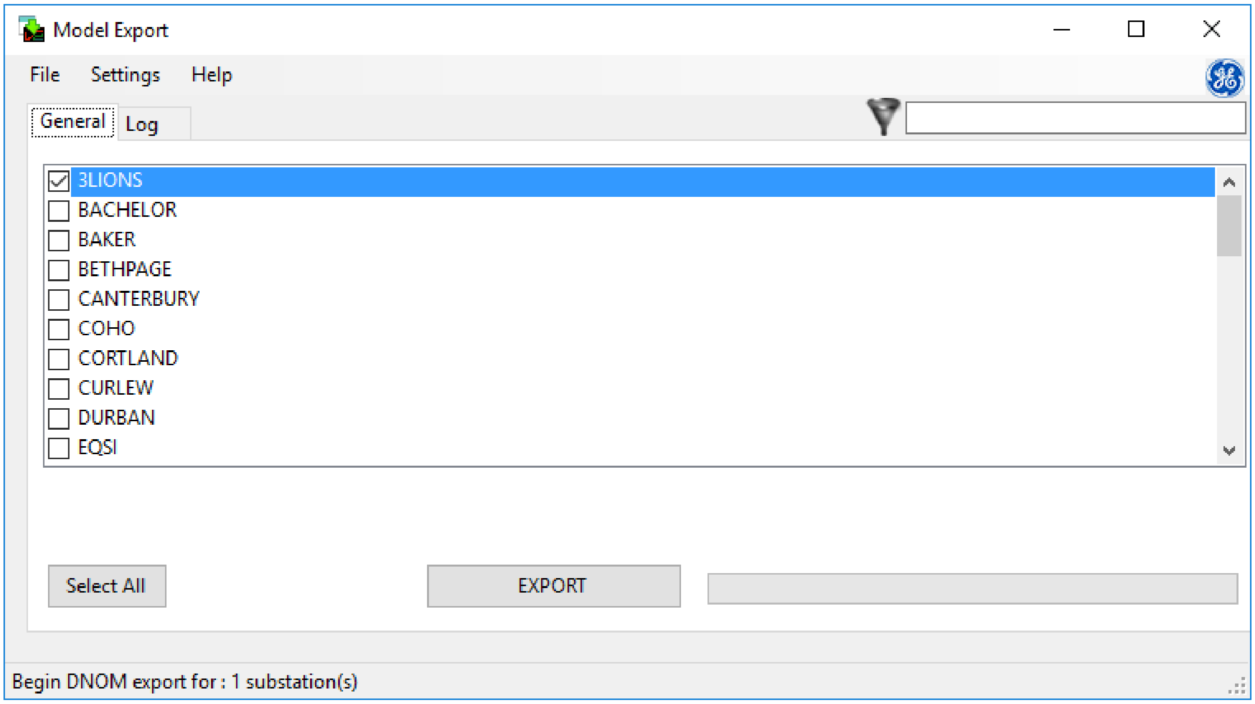


Figure 4. General Tab

The fields and controls on the General tab include:

- **Filter:** Filters the stations list by the specified filtering criteria.
- **Select:** Selects or deselects a station.
- **Select All/Clear All:** Selects all stations in the list or cancels the selection for all stations.
- **Export:** Starts the export process for the selected stations.
- **Status bar:** The status bar at the bottom of the display shows the tool's status.

3.3. Log Tab

The Log tab shows the log of the data extraction process for the specified station.

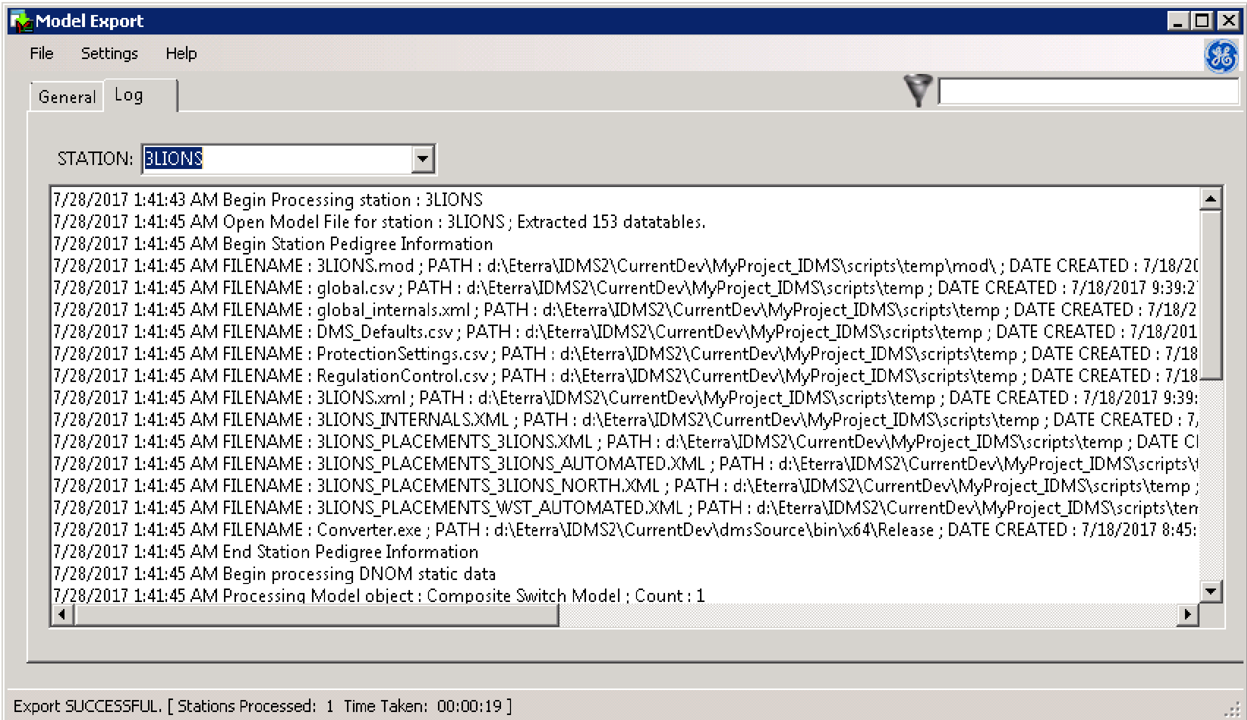


Figure 5. Log Tab

3.4. Configuration

On the Configuration display, you can configure Model Export.

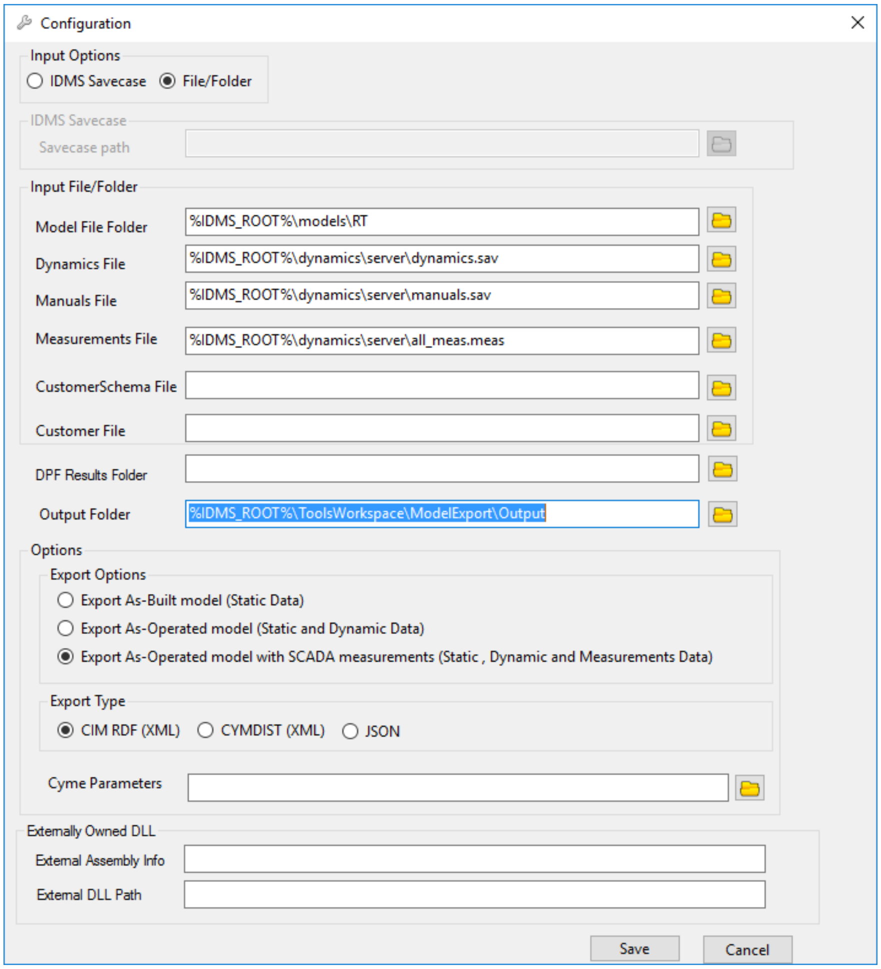


Figure 6. Model Export Configuration

The Configuration display contains the following:

- **Input Options:** Specifies the type of input file(s) to be used by Model Export.
    - **ADMS Savecase:** Select this option to extract data from a savecase.
    - **File/Folder:** Select this option to extract data from models, dynamics, manuals, and SCADA measurement files.
- **Savecase Path:** Path to the DMS savecase to be extracted. This option is available only when the ADMS Savecase input option is selected.
- **Input File/Folder:** Paths to input files and folders with files. The File/Folder options are available only when the File/Folder input option is selected.
    - **Model File Folder:** Path to the folder with station model files (*.mod).
    - **Dynamics File:** Path to the dynamics file (by default, `dynamics.sav`).
    - **Manuals File:** Path to the file with manual entries, such as data entered via data entry pop-ups or manual input tabulars (by default, `manuals.sav`).
    - **Measurements File:** Path to the file with measurements (by default, all_meas.meas).

    - **Customer File:** Path to the file (*.txt) with customer data that is used to build the customer model. For more information, refer to the Customer List Compiler User Guide.
    - **Schema file:** Path to the file (*_customerSchema.xml) with the schema that is used to interpret customer data in the customer file.
- **Output Folder:** Path to the folder where Model Export extracts data and records logs.
- **Export Options:** Options to specify the data types to be extracted.
    - **Export As-Built Model (Static Data):** If this option is selected, Model Export exports static data from the stations model files (*.mod).
    - **Export As-Operated Model (Static Data and Dynamic Data):** If this option is selected, Model Export exports both static data and dynamic data. Dynamic data is exported from the dynamics file (dynamics.sav).
    - **Export As-Operated Model with SCADA Measurements (Static Data, Dynamic Data, and Measurements Data):** If this option is selected, Model Export exports static data, dynamic data, and SCADA measurement data. SCADA measurement data is exported from the measurement files (*.nmea).
- **Export Options:** Options to specify the type of output file.
    - **CIM RDF (XML):** If this option is selected, Model Export exports static data from the stations model files (*.mod) in the CIM format.
    - **CYMDIST (XML):** If this option is selected, Model Export exports static data from the stations model files (*.mod) in the CYMDIST format used by the Eaton's CYME software.
    - **JSON:** If this option is selected, Model Export exports static data from the stations model files (*.mod) in the JSON format.
- **CYME Parameters:** Path to the file that defines string parameters to be used in the CYME export file. Used for the CYMDIST export type.
- **Externally Owned DLL:** This section specifies parameters for converting system coordinates (X, Y) into projected coordinate system (PCS) coordinates (latitude, longitude). Used for the JSON export type.
    - **External Assembly Info:** Name of the class that converts system X/Y coordinates into latitude and longitude coordinates. The value format is namespace.class. For example, the demo coordinate converter class is SampleCoordinateConverter.CoordinateConverter.
    - **External DLL Path:** The full path to the assembly (DLL file) that contains a class for converting system X/Y coordinates into latitude and longitude coordinates. For example, the path to the assembly with the demo coordinate converter class, is `D:\Eterra\Distribution\Client\bin\NetUtil.dll`.

> **Note**
>
> For US customers, the demo coordinate converter can be used. For non-US customers, a custom converter must be developed.

4. Using Model Export

This chapter describes how to use the general functions of the Model Export tool.

4.1. Launching the Model Export

To open the Model Export, from the Windows Start menu, navigate to Eterra > Distribution > Model Export > Model Export.

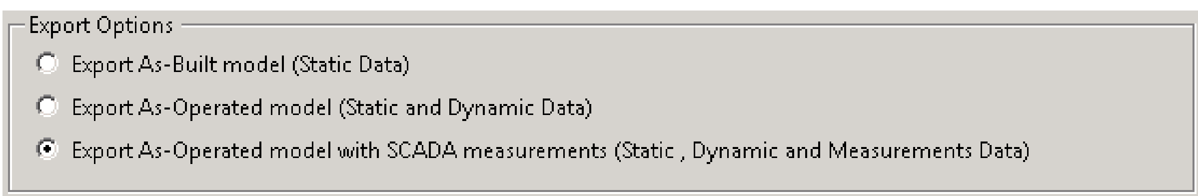
Typically, the Model Export configuration file is provided in the Target field of the Model Export shortcut. In this case, the tool is launched with the specified configuration file.

If the configuration file is not already provided in the shortcut's Target field, you may need to browse for and open the configuration file manually.

4.2. Extracting Data

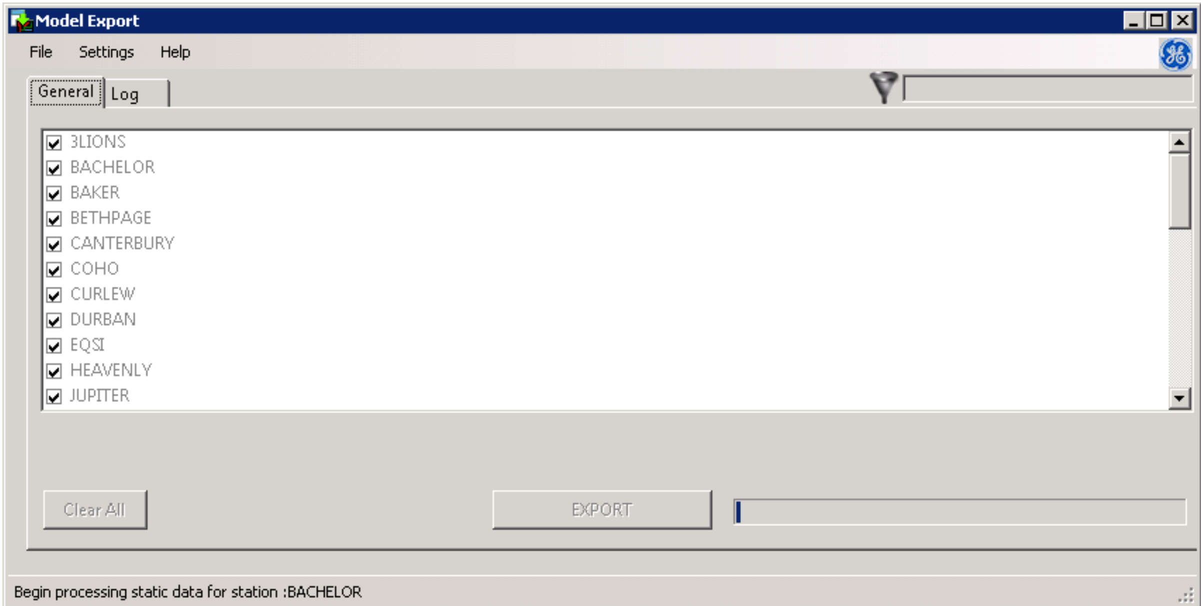
To extract data in the CIM format:

1. Configure the Model Export to extract the required types of data (static data, dynamic data, and SCADA measurements), as described in Configuration.

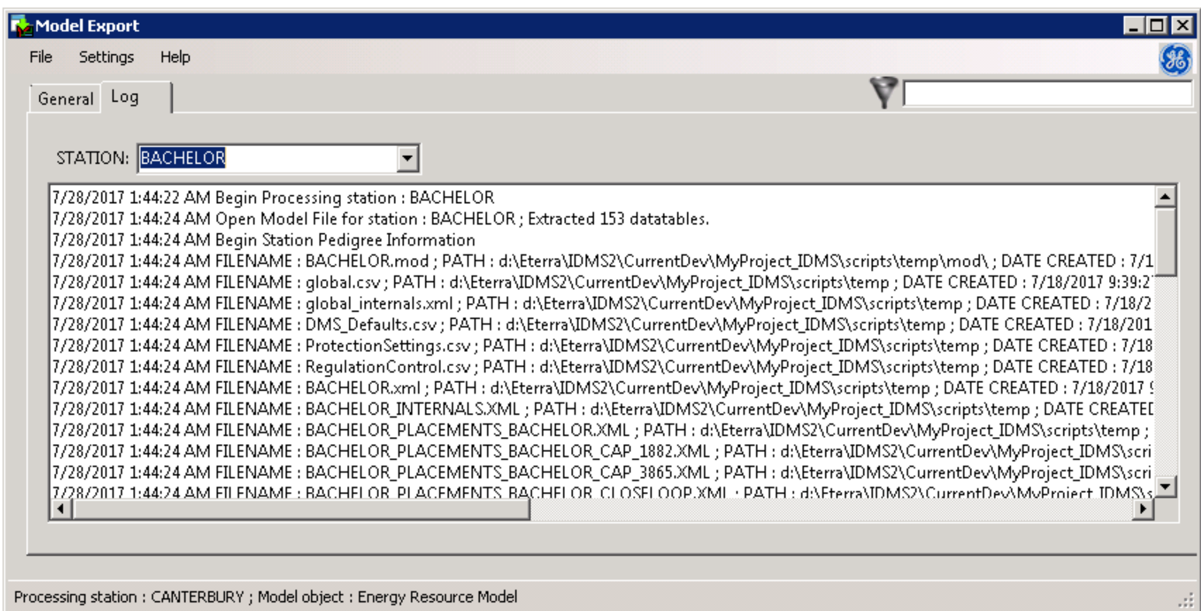
   

2. On the General tab, select the stations and then click Export.

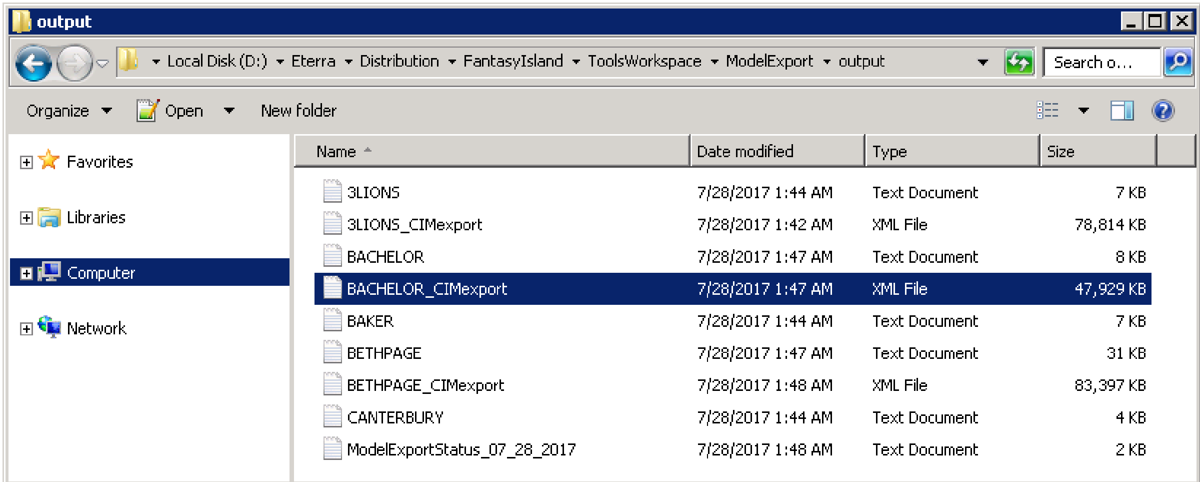
   The export process starts, and the status is shown on the status bar.

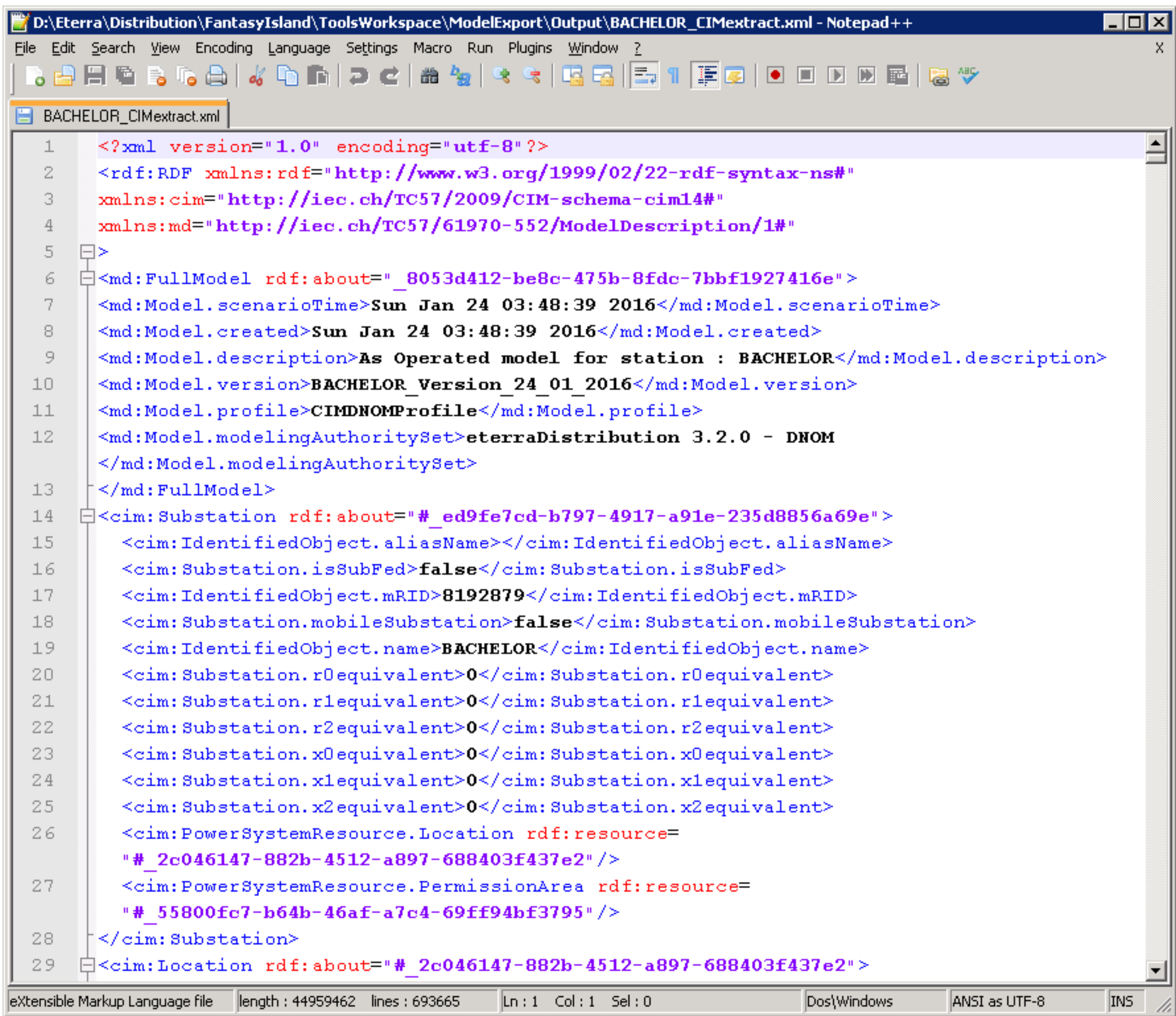


3. Wait until the export process is completed.
4. On the Log tab, check the export log for stations.



5. Navigate to the Output folder and check the generated CIM extract files.


output
Local Disk (D:) ▸ Eterra ▸ Distribution ▸ FantasyIsland ▸ ToolsWorkspace ▸ ModelExport ▸ output
Search o...
Organize
Open
New folder
Favorites
Libraries
Computer
Network
Name
Date modified
Type
Size
3LIONS 7/28/2017 1:44 AM Text Document 7 KB
3LIONS_CIMexport 7/28/2017 1:42 AM XML File 78,814 KB
BACHELOR 7/28/2017 1:47 AM Text Document 8 KB
BACHELOR_CIMexport 7/28/2017 1:47 AM XML File 47,929 KB
BAKER 7/28/2017 1:44 AM Text Document 7 KB
BETHPAGE 7/28/2017 1:47 AM Text Document 31 KB
BETHPAGE_CIMexport 7/28/2017 1:48 AM XML File 83,397 KB
CANTERBURY 7/28/2017 1:44 AM Text Document 4 KB
ModelExportStatus_07_28_2017 7/28/2017 1:48 AM Text Document 2 KB


D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\ModelExport\Output\BACHELOR_CIMextract.xml - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window ?
BACHELOR_CIMextract.xml
<?xml version="1.0" encoding="utf-8"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
xmlns:cim="http://iec.ch/TC57/2009/CIM-schema-cim14#"
xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#"
>
<md:FullModel rdf:about="_8053d412-be8c-475b-8fdc-7bbf1927416e">
<md:Model.scenarioTime>Sun Jan 24 03:48:39 2016</md:Model.scenarioTime>
<md:Model.created>Sun Jan 24 03:48:39 2016</md:Model.created>
<md:Model.description>As Operated model for station : BACHELOR</md:Model.description>
<md:Model.version>BACHELOR_Version_24_01_2016</md:Model.version>
<md:Model.profile>CIMDNOMProfile</md:Model.profile>
<md:Model.modelingAuthoritySet>eterraDistribution 3.2.0 - DNOM
</md:Model.modelingAuthoritySet>
</md:FullModel>
<cim:Substation rdf:about="#_ed9fe7cd-b797-4917-a91e-235d8856a69e">
<cim:IdentifiedObject.aliasName></cim:IdentifiedObject.aliasName>
<cim:Substation.isSubFed>false</cim:Substation.isSubFed>
<cim:IdentifiedObject.mRID>8192879</cim:IdentifiedObject.mRID>
<cim:Substation.mobileSubstation>false</cim:Substation.mobileSubstation>
<cim:IdentifiedObject.name>BACHELOR</cim:IdentifiedObject.name>
<cim:Substation.r0equivalent>0</cim:Substation.r0equivalent>
<cim:Substation.r1equivalent>0</cim:Substation.r1equivalent>
<cim:Substation.r2equivalent>0</cim:Substation.r2equivalent>
<cim:Substation.x0equivalent>0</cim:Substation.x0equivalent>
<cim:Substation.x1equivalent>0</cim:Substation.x1equivalent>
<cim:Substation.x2equivalent>0</cim:Substation.x2equivalent>
<cim:PowerSystemResource.Location rdf:resource=
"#_2c046147-882b-4512-a897-688403f437e2"/>
<cim:PowerSystemResource.PermissionArea rdf:resource=
"#_55800fc7-b64b-46af-a7c4-69ff94bf3795"/>
</cim:Substation>
<cim:Location rdf:about="#_2c046147-882b-4512-a897-688403f437e2">
eXtensible Markup Language file
length : 44959462 lines : 693665
Ln : 1 Col : 1 Sel : 0
Dos\Windows
ANSI as UTF-8
INS

4.3. Command-Line Interface

The Model Export's command-line interface supports the following options.

/CONFIG [*PATH TO FILE*]

Specifies the path to the Model Export configuration file. All export settings, including Input and Output folders and export options, are taken from this configuration file.

/TARGET [*XXXX*]

Specifies the stations to create CIM models for. The following arguments can be used:

- **/TARGET [StationName1,StationName2,StationName3]**

  Specifies the station names to be processed by Model Export in one string. Station names are separated with commas.

  For example:

  ```
  /TARGET "3LIONS,BACHELOR,RAVEN,PEBBLE BEACH"
  ```

- **/TARGET ***

  Specifies all stations in the input folder for extracting.

- **/TARGET [_TXTFilePath]**

  Specifies all stations in the TXT file for conversion. It provides the path to the TXT file, including the file name. The TXT file lists all of the stations to be processed by Model Export, with one station per line.

  For example:

  ```
  /TARGET D:\eterra\Distribution\FantasyIsland\ToolsWorkspace\Model Export\input\StationNames.txt
  ```

/HELP

Shows the help for command line options.

/BATCH

If this parameter is used, the Model Export tool is working in silent mode (without user interface).

The following example creates CIM files for all stations stored in the Input folder, including static data, dynamics data, and SCADA measurements, as specified in the configuration file:

```
d:\Eterra\Distribution\ModelExport\Bin\ModelExport.exe /CONFIG
d:\Eterra\Distribution\ModelExport\Bin\ModelExportConfig.xml /Target * /Batch
```

5. Example of a CIM Extract

This chapter provides part of the CIM extract data for a substation object.

```
<cim:Substation rdf:about="#_c630ae04-edc9-45d5-b94e-fa634d5b5232">
  <cim:IdentifiedObject.aliasName></cim:IdentifiedObject.aliasName>
  <cim:Substation.isSubFed>false</cim:Substation.isSubFed>
  <cim:IdentifiedObject.mRID>8214381</cim:IdentifiedObject.mRID>
  <cim:Substation.mobileSubstation>false</cim:Substation.mobileSubstation>
  <cim:IdentifiedObject.name>PINE VALLEY</cim:IdentifiedObject.name>
  <cim:Substation.r0equivalent>0</cim:Substation.r0equivalent>
  <cim:Substation.r1equivalent>0</cim:Substation.r1equivalent>
  <cim:Substation.r2equivalent>0</cim:Substation.r2equivalent>
  <cim:Substation.x0equivalent>0</cim:Substation.x0equivalent>
  <cim:Substation.x1equivalent>0</cim:Substation.x1equivalent>
  <cim:Substation.x2equivalent>0</cim:Substation.x2equivalent>
  <cim:PowerSystemResource.Location rdf:resource="#_5ac16e28-a458-467b-a7ec-2f8da5d4574e"/>
  <cim:PowerSystemResource.PermissionArea rdf:resource="#_9a536635-76da-4cb4-910c-39b3d89a00d4"/>
</cim:Substation>
<cim:Location rdf:about="#_5ac16e28-a458-467b-a7ec-2f8da5d4574e">
  <cim:IdentifiedObject.aliasName></cim:IdentifiedObject.aliasName>
  <cim:Location.direction></cim:Location.direction>
  <cim:Location.electronicAddress></cim:Location.electronicAddress>
  <cim:Location.geoInfoReference></cim:Location.geoInfoReference>
  <cim:IdentifiedObject.mRID>8214381:L</cim:IdentifiedObject.mRID>
  <cim:Location.mainAddress>1111 Main St PINE VALLEY</cim:Location.mainAddress>
  <cim:IdentifiedObject.name></cim:IdentifiedObject.name>
  <cim:Location.phone1></cim:Location.phone1>
  <cim:Location.phone2></cim:Location.phone2>
  <cim:Location.secondaryAddress></cim:Location.secondaryAddress>
  <cim:Location.status></cim:Location.status>
  <cim:Location.type></cim:Location.type>
  <cim:Location.CoordinateSystem rdf:resource="#_371fc6f4-d9f0-4fe0-ab1a-c16b703128fe"/>
</cim:Location>
<cim:PermissionArea rdf:about="#_9a536635-76da-4cb4-910c-39b3d89a00d4">
  <cim:IdentifiedObject.aliasName></cim:IdentifiedObject.aliasName>
  <cim:IdentifiedObject.mRID>8214381:PA</cim:IdentifiedObject.mRID>
  <cim:IdentifiedObject.name>PINEVA</cim:IdentifiedObject.name>
</cim:PermissionArea>
```